

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Machined Block with 2 Holes

Commodity: MACHINING · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 6 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥74.17	3.8%	56.5%	20.1%	7.4%

Model Confidence: **Low** · 6 traced operations · 14 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 44.9 cm³; mass for the selected material family 0.352 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: machining

Alloy / material: 1045 / C45 / 080M46 (Medium-Carbon Steel) · Net weight: 0.352 kg (measured)

Material utilisation: 93% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 6

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥2.81	3.8%	
2. Process (Machine)	¥41.88	56.5%	
3. Direct Labour	¥14.90	20.1%	
4. Tooling (amortised)	¥0.68	0.9%	
5. Packaging	¥0.54	0.7%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥0.63	0.9%	
Factory Cost	¥61.44	82.8%	
7. Overhead (SG&A)	¥7.23	9.8%	
Subtotal	¥68.67	92.6%	
8. Supplier Margin	¥5.49	7.4%	
TOTAL SHOULD-COST	¥74.17	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥0.54	Logistics / part	¥0.63
Total Tooling Cost	¥135746.61	Amortisation Volume	200,000 parts

Geometric DFM / DFA - measured from the CAD model

0 issue(s) across 0 instance(s), from 7 measured feature(s) and 3 rule(s). Every issue names the B-rep faces that produced it and the published source of its threshold. No finding on this part has a modelled cost path; each says why.

No geometric findings

No rule in the pack fired on the measured features. Read this alongside the limitations below - it means the checks that RAN found nothing, not that the part is clean.

DFA - handling & insertion (Boothroyd method, geometric half)

Handling 1.63s + insertion 1.5s = 3.13s, 1.04x the 3s ideal part.

+0.5s Asymmetric envelope — the part must be oriented before insertion (bounding box 60×40×20 mm, no two dimensions within 5%)

Theoretical minimum part count, and therefore the Boothroyd design-efficiency index, requires the full assembly — this analysis covers one part, so no efficiency ratio is given.

Nesting and tangling risk is not assessed: it depends on how parts are presented in bulk, which the geometry alone does not determine.

Rotational symmetry is approximated from the bounding box, not from an inertia-tensor shape comparison.

What was NOT checked

solid-bodied part (fill ratio 0.9346) — single-ray thickness measures part extent, not wall; section-change and hot-spot rules suppressed 2 of 7 per-face thickness readings discarded as ray artefacts — outside 2.76–27.63 mm, i.e. 0.3x to 3x the characteristic 2·V/S thickness of 9.21 mm. Readings below the floor are rays grazing a face; readings above the cap are rays crossing the part. Wall-derived checks ran only on the survivors.

No material family was confirmed, so any check whose threshold depends on the alloy or resin did not run.

Tolerance and surface-finish callouts were not checked — they are on the drawing, not in the solid, and they drive whether a feature is milled, ground or honed.

Tool access and fixturing were not simulated: a feature can be geometrically fine and still unreachable in the chosen setup.

Thread specifications were not verified against standard tap sizes.

Workholding and part distortion under clamping were not assessed.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-steel1045	ID	
Grade / Specification	1045 / C45 / 080M46 (Medium-Carbon Steel)		UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	0.3520 kg	kg	Weight in finished part
Gross Weight (stock)	0.3769 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	0.0249 kg	kg	Gross - Net
Material Utilisation	93.4%	%	Benchmark: machining 60–75 % (stock removal)
Material Price	¥7.57	CNY /kg	UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.880 (millSteel)
Scrap Recovery Price	¥1.75	CNY /kg	
Gross Material Cost	¥2.85	CNY	Gross × price/kg
Scrap Credit	-¥0.04	CNY	Scrap × recovery price
NET RAW MATERIAL COST	¥2.81	CNY	Gross - scrap credit + consumables
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Setup (amortised)	HAAS VF-2 (3-axis Mill)	¥212.31	1.20	1	100.0 %	¥4.25
Milling — +X (4 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	2.78	1	85.0%	¥11.57
Milling — +Y (2 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	1.40	1	85.0%	¥5.83
Milling — +Z (2 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	1.40	1	85.0%	¥5.83
Drilling — 2 holes	CNC Drilling	¥136.82	3.82	1	85.0%	¥10.25

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
(2×Ø10.0×20) [geometry-measured]	Centre					
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured]	HAAS VF-2 (3-axis Mill)	¥212.31	1.00	1	85.0%	¥4.16
TOTAL						¥41.88

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Setup (amortised)	Skilled Machinist	¥71.49	1	1.20	100.0%	¥1.43	¥5.68
Milling — +X (4 faces)	Skilled Machinist	¥71.49	1	2.78	92.0%	¥3.60	¥15.16
Milling — +Y (2 faces)	Skilled Machinist	¥71.49	1	1.40	92.0%	¥1.81	¥7.64
Milling — +Z (2 faces)	Skilled Machinist	¥71.49	1	1.40	92.0%	¥1.81	¥7.64
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured]	Skilled Machinist	¥71.49	1	3.82	92.0%	¥4.95	¥15.21
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured]	Skilled Machinist	¥71.49	1	1.00	92.0%	¥1.30	¥5.46
TOTAL						¥14.90	¥56.78

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø10.0 × 20 thru	2	Drilling	1.10	¥4.66	Yes
Face	2243 mm²	2	Face milling (facing)	0.96	¥4.07	Yes
Face	1200 mm²	2	Face milling (facing)	0.70	¥2.97	Yes
Face	800 mm²	2	Face milling (facing)	0.60	¥2.54	Yes
TOTAL costed		8		3.36	¥14.25	Yes

§5 — Machine Rate Buildup

CNC Drilling Centre [mach-drill] · Confidence: Low · UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥139366.52	¥43.55	4,000 hr/yr available
Maintenance	¥59728.51	¥18.67	
Energy	¥49577.02	¥15.49	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥79638.01	¥24.89	
Finance Cost	¥69683.26	¥21.78	
TOTAL — CNC Drilling Centre	¥437812.32	¥136.82	Effective hours: 3200/yr

HAAS VF-2 (3-axis Mill) [mach-haas-vf2] · Confidence: Low · HAAS VF-2 benchmark, UK Tier-2 machining shop | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @50% util	Notes
Depreciation	¥189140.27	¥94.57	4,000 hr/yr available
Maintenance	¥89592.76	¥44.80	
Energy	¥28919.93	¥14.46	
Floor Space	¥24886.88	¥12.44	
Indirect Support	¥47285.07	¥23.64	
Finance Cost	¥44796.38	¥22.40	
TOTAL — HAAS VF-2 (3-axis Mill)	¥424621.29	¥212.31	Effective hours: 2000/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf
material.pricePerKg	0.8360	£/kg	UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.880 (millSteel)	mat-steel1045	Medium

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.scrapRecoveryPricePerKg	0.1936	£/kg	UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.880 (millSteel)	mat-steel1045	Medium
Setup (amortised).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Setup (amortised).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +X (4 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +X (4 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +Y (2 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +Y (2 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +Z (2 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +Z (2 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured].machineRatePerHr	15.1182	£/hr	UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-drill	Low
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured].machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥74.17 ± 20.2%	¥60.18	¥73.30	¥90.05	wide	16.1%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 12 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Setup (amortised): Machine Rate	¥212.31/hr	-5.2%	+5.2%	¥7.65
Milling — +X (4 faces): Machine Rate	¥212.31/hr	-5.2%	+5.2%	¥7.65
Milling — +Y (2 faces): Machine Rate	¥212.31/hr	-5.2%	+5.2%	¥7.65
Milling — +Z (2 faces): Machine Rate	¥212.31/hr	-5.2%	+5.2%	¥7.65
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured]: Machine Rate	¥212.31/hr	-5.2%	+5.2%	¥7.65
Drilling — 2 holes (2×Ø10.0×20) [geometry-measured]: Cycle Time	0.06hr	-2.5%	+2.5%	¥3.68
Milling — +X (4 faces): Cycle Time	0.05hr	-2.5%	+2.5%	¥3.67
Setup (amortised): Labour Rate	¥71.49/hr	-2.4%	+2.4%	¥3.60

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥3.19	¥76.15	¥53.64	¥1.23	¥9.64	¥143.85	¥0.44	¥150.56	+103%
Germany (EUR)	¥3.29	¥79.95	¥86.69	¥1.30	¥13.53	¥184.75	¥0.50	¥193.12	+160%
France (EUR)	¥3.25	¥70.06	¥62.31	¥1.14	¥10.32	¥147.07	¥0.50	¥154.00	+108%
Spain (EUR)	¥3.19	¥67.01	¥39.28	¥1.09	¥7.55	¥118.11	¥0.50	¥123.93	+67%
Poland (PLN)	¥3.09	¥54.83	¥24.38	¥0.89	¥5.08	¥88.27	¥0.52	¥92.88	+25%
Turkey (TRY)	¥2.87	¥45.69	¥13.54	¥0.74	¥3.52	¥66.37	¥0.57	¥70.11	-5%
China (CNY)	¥2.81	¥41.88	¥14.90	¥0.68	¥7.23	¥67.50	¥0.63	¥74.17	Base
India (INR)	¥2.87	¥39.60	¥9.48	¥0.64	¥2.72	¥55.31	¥0.66	¥58.61	-21%
Mexico (MXN)	¥3.03	¥45.69	¥15.71	¥0.74	¥3.65	¥68.82	¥0.59	¥72.65	-2%
United States (USD)	¥3.19	¥64.73	¥70.43	¥1.05	¥9.51	¥148.91	¥0.55	¥155.88	+110%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

1.09 kgCO2e · 3.11 /kg · cradle-to-gate

Material	0.75 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	0.29 kgCO2e	Grid intensity	0.550 kg/kWh · 0.53 kWh
Logistics	0.06 kgCO2e	Total	1.09 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 56.5% of total is above the benchmark range of 18–35%. Cycle time or machine rate is elevated.
Impact	High · up to 11% potential saving
Benchmark	Process %: yours 56.5% vs industry 18–35%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixturing time

[INFO] Material cost is below benchmark range

Finding	Material at 3.8% is below the typical 30–55% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 3.8% vs industry 30–55%
Action 1	Focus optimisation efforts on process and labour efficiency

[INFO] Routing already consolidated — no multi-axis lever left

Finding	5 machining operation(s) run across 2 station(s). The consolidation a reviewer would suggest is already assumed in this cost — quote it as evidence, not as an action.
Impact	Low
Action 1	Use the routing basis in the derivation trace to defend the process cost line

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥8.68/part (~12%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Process & Automation	Consider near-net-shape pre-form (casting/forging)	Process cost dominates (>40%) — Process cost at 56.5% of total. Machining conversion cost is unusually high.	¥5.93	8.0%	Medium Term	Medium
2	Process & Automation	Unmanned / Lights-Out Running	Labour is 20.1% of cost on a machine-paced part with healthy OEE (88%) — the classic profile for bar-feed or pallet-pool unattended running.	¥4.45	6.0%	Medium Term	Medium
3	Process & Automation	Multi-Machine Manning	Operations are manned at up to 1.0 operators per machine while the machine paces the cycle. One operator tending two machines is	¥3.71	5.0%	Quick Win	Low

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
			the textbook correction.				
4	Design & Geometry	Part-Count / Feature Integration Review (Boothroyd)	6 operations suggest features or joined parts that may not need to exist separately. The Boothroyd minimum-part-count test asks of each: does it move, must it be a different material, must it be separate for assembly?	¥4.45	6.0%	Long Term	Medium
5	Design & Geometry	Tolerance and Surface-Finish Relaxation	Machining is 56.5% of the part. The tightest tolerance and finish callouts set the machine, the cycle and the inspection burden — reviewing them is the cheapest cost lever that exists.	¥2.97	4.0%	Quick Win	Low
6	Commercial & Sourcing	Learning-Curve Price-Down Agreement	Labour is 20.1% of cost and no learning curve is priced in. On a multi-year programme, labour productivity improves whether or not the contract captures it — the contract should.	¥2.23	3.0%	Quick Win	Low
7	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥2.23	3.0%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	3	Consider near-net-shape pre-form (casting/forging)	¥5.93	¥14.09
Design & Geometry	2	Part-Count / Feature Integration Review (Boothroyd)	¥4.45	¥7.42
Commercial & Sourcing	2	Learning-Curve Price-Down Agreement	¥2.23	¥4.45

§14 — Inputs to Confirm

No saving is claimed against these

Item	What the costing assumed	How to close it
Multi-operation routing already consolidated	5 operations run across 2 station(s) — the setup-reduction lever is already taken in this cost.	Use the routing comparison in the derivation trace as negotiation evidence

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Multi-Machine Manning
- Tolerance and Surface-Finish Relaxation
- Learning-Curve Price-Down Agreement
- Annual Volume Re-Commitment for Better Pricing

Long-Term Changes — Strategic Investment

- Part-Count / Feature Integration Review (Boothroyd)